Unit: Unit 4: Absolute Value
Topic: Solving Absolute Value Equations

Name: ____________
Date: ________

Quiz (Habanero)

Q1. If $|x| = 9$, what are the possible values of x?
(A) $x = 9$
(B) $x = -9$
(C) $x = 9$ or $x = -9$
(D) $x = 0$
(E) $x = 18$

Q2. Solve for x: $|2x| = 12$
(A) $x = 5$
(B) $x = 6$
(C) $x = 5$ or $x = -6$
(D) $x = 3$
(E) $x = 4$

Q3. What is the solution to $|x - 2| = 7$?
(A) $x = 5$
(B) $x = 9$
(C) $x = 5$ or $x = -5$
(D) $x = 9$ or $x = -5$
(E) $x = 4$

Q4. The temperature in a region is expected to rise by $|3x|$ degrees over the next week. If the temperature is expected to rise by 12 degrees, what is the value of x?
(A) $x = 2$
(B) $x = 4$
(C) $x = 2$ or $x = -4$
(D) $x = 6$
(E) $x = 8$

Q5. A weather forecast predicts $|2x-5|$ inches of rainfall. If the forecast predicts 11 inches of rainfall, what is the value of x?
(A) $x = 3$
(B) $x = 8$
(C) $x = 3$ or $x = 8$
(D) $x = 2$
(E) $x = 9$

Q6. Solve for x: $|x + 1| = |2x - 3|$
(A) $x = 2$
(B) $x = 4$
(C) $x = 2$ or $x = -1$
(D) $x = 1$
(E) $x = 3$

Q7. If $|x| = |2x - 1|$, what are the possible values of x?
(A) $x =$
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(A) $x = 1$
(B) $x = 0$ or $x = 1$
(C) $x =$
rac12orx $= 1$
(C) $x = 0$

Q8. The depth of water in a lake is $|x - 10|$ feet. If the depth of water is 5 feet, what is the value of x?
(A) $x = 15$
(B) $x = 5$
(C) $x = 15$ or $x = 5$
(D) $x = 20$
(E) $x = 10$

Open Ended Questions (FRQ)

Q9. A plant's growth is modeled by the equation $|2x - 5| = 11$. Solve for x and explain your reasoning.

Q10. The temperature difference between two cities is $|3x + 2|$ degrees. If the temperature difference is 14 degrees, solve for x and check your solution for extraneous results.

Chef's Tips

- Tip 1: Ensure students check for extraneous solutions
- Tip 2: Remind students to consider both positive and negative solutions for absolute value equations
- Tip 3: Encourage students to use algebraic methods to solve absolute value equations